

ABBOTT NUTRITION

Supply Chain Strategy Analysis

OM725 – Spring 2026 Team Project

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Abbott Nutrition: Overview

\$8.5B

Nutrition Revenue
(2025)

160+

Countries

36

U.S. State
WIC Contracts

13+

Owned Global
Mfg Plants

Industry & Products

- Nutrition division of Abbott Laboratories (NYSE: ABT).
- 19% of Abbott's total revenue; leading U.S. pediatric & adult nutrition company.
- Product lines: Infant formula, adult/pediatric oral nutrition, enteral/clinical nutrition.



Markets & Competitors

- U.S. infant formula: \$6.5B, concentrated oligopoly (90% held by 4 firms).
- Global clinical/adult nutrition: \$34B, 5.5% CAGR; Abbott holds 20-26% share.
- Infant formula competitors: Mead Johnson / Reckitt (Enfamil), Perrigo (store brands).

Business Strategy: How Abbott Competes

Abbott Nutrition competes on science-based differentiation, not price.



Science-Based Innovation

700+ scientists globally. Stage-Gate R&D. Clinically validated products command premium pricing (e.g., Ensure, EleCare).



Institutional Entrenchment

Similac is #1 in U.S. hospitals. HCP recommendations at birth drive long-term brand loyalty regardless of WIC status.



Global Expansion (Adult)

Co-located R&D + manufacturing in Singapore, India, Ireland, NZ. Adult Nutrition grew 11.4% organically in 2024.



WIC Government Lock-In

Sole-source contracts in 36 states + D.C. produce ~6x market share spillover. New contract through 2030.

Abbott's PSMD / Q Analysis

Element	Configuration	Key Details
Planning	Make-to-Stock (MTS)	Demand-forecast driven. SAP ERP + Planful across 60+ country affiliates. Diverse product portfolios in continuous replenishment. JIT principles within MTS (quick changeovers, surge tanks).
Sourcing	Primarily Commodities (w/ pharma-grade specs)	Dairy proteins, corn maltodextrin, vegetable oils, mineral premixes - global commodities. But Abbott imposes pharmaceutical-grade quality standards (FDA cGMP, SQF, Cronobacter testing at 6× federal minimum).
Manufacturing	Self-Manufactured (13+ owned plants)	Core products never outsourced. Specialty/metabolic formulas ONLY made in-house. 3rd-party used for overflow only. Aseptic filling at Tipp City (1M bottles/day). Global footprint: US, Ireland, Singapore, China, India, Spain.
Distribution	Centralized Procurement Decentralized Execution	4 channels: Hospital (direct), WIC (gov't contract), Retail (3PL/wholesale), International (regional hubs - Singapore for APAC, Ireland for EU/LATAM). Products in mass retail, drug, grocery, military, convenience.
Quality	Design + Inspect (heavily regulated)	Design-in: Aseptic processing preserves nutrients, reduces contamination risk. Inspect-in: 4-step finished product release, 200+ checks/shift, 45-point inspection, SQF Level III. Post-2022: \$60M Sturgis upgrades, \$536M new Bowling Green plant.

Supply Chain as Competitive Advantage

SC Element	Role	Why?
Delivery Reliability (D)	ORDER WINNER #1	WIC contracts in 36 states require massive delivery scale. Winning a state's WIC = 6× market share spillover. No smaller competitor can match this infrastructure.
Product Mix / Flexibility (M)	ORDER WINNER #2	4,000+ SKUs. Sole U.S. source for specialty metabolic formulas (5,000 children). Hospital formulary = structural switching costs. Lifecycle coverage from infant to geriatric.
Quality (Q)	ORDER QUALIFIER	All FDA-approved formulas are safe. Quality compliance is mandatory but doesn't differentiate. Exception: specialty metabolic formulas where quality IS the winner.
Price (P)	ORDER QUALIFIER	Deep WIC rebates (79–90% of wholesale) are required. All 3 major manufacturers offer similar rebates. The FTC opened a collusion investigation in 2023 into whether this similarity reflects coordination among the 3 firms holding 70 WIC contracts. Price is necessary but not sufficient to win.



Key Insight: Abbott's order winners are Delivery and Mix, not price or quality. The WIC program structure makes delivery reliability at massive scale literally worth billions, and product breadth creates switching costs that lock in hospitals.

Lean & Six Sigma at Abbott Nutrition

✓ Lean Implementation

"Our quality and operations teams use Lean Six Sigma and Class A programs to accomplish productivity improvements" - Abbott Nutrition VP of Operations. 200+ quality checks/shift; rapid-changeover lines; Casa Grande AZ earned SQF Level III certification.

✓ Six Sigma: DMAIC

Abbott deploys DMAIC via certified Black Belt & Green Belt practitioners on cross-functional projects. Formal Lean/Six Sigma Programs Manager role maintained. Staff certified from White Belt to Master Black Belt via external programs.

✓ Lean Six Sigma in Practice

Steering committee uses case-study approach to transfer solutions across all sites. Documented example: time-lapse video on production floor identified operator workstation orientation as waste source, directly reducing equipment-stoppage lag times.

✓ Lean Results: Sligo

Batch right-first-time rate >99% since 2014. 85% reduction in CAPA non-conformity reports since 2013. Zero-waste-to-landfill status achieved. Measurable, documented outcomes of Lean Six Sigma discipline in Abbott Nutrition manufacturing.

✓ 5 Shingo Prizes Won

Most active Shingo Prizes of any company globally (2019). Abbott Nutrition recipients: Sligo Ireland (2017), Granada Spain (2018), Singapore (2019), Sturgis MI (2019), Jhagadia India (2022). Each prize requires independent on-site assessment by examiners.

✓ Digital Lean

Abbott Longford built a formal Lean Management System (LMS) aligned with Shingo principles. Granada factory layers AVEVA PI System for real-time vertical data integration and a Manufacturing Unified Name Space onto its Lean Six Sigma foundation.

Differentiating Supply Chain Tactics

✓ Aseptic Processing

Tipp City: 1M bottles/day. Extends shelf life, reduces spoilage, expands geographic reach — rare cost + quality win-win.

✓ Post-2022 Diversification

Reduced Sturgis production 41%. \$536M Bowling Green plant for specialty formulas. Eliminates single-source risk.

✓ Co-Located R&D + Mfg

Singapore, Shanghai, Bengaluru, UIUC. Local scientists identify preferences → pilot plants develop → manufacturing scales without intercontinental delays.

✓ Centralized Procurement / Decentralized Execution

Global purchasing power with local market knowledge. Supplier diversity: \$1B+ annual diverse spend (Billion Dollar Roundtable).

✓ WIC Cross-Portfolio Margin

Adult Nutrition cross-subsidizes deep WIC rebates (79–90% off wholesale). Pure-play infant formula companies cannot replicate.

✓ Global Operations Council

Senior leadership governance for resilience: sourcing continuity plans, regular supplier audits. Enabled COVID-19 test production while maintaining nutrition supply.

THANK YOU

ANY QUESTIONS?

Key Takeaways

- 1 Abbott competes on science-based differentiation not price but across the full nutrition lifecycle from infant to geriatric.
- 2 Delivery reliability at WIC scale is the #1 order winner: 36 state contracts lock in 52% of formula-fed WIC infants with massive spillover into non-WIC market share.
- 3 Product mix/flexibility is order winner #2: 4,000+ SKUs, sole-source metabolic formulas, and full hospital formulary coverage create structural switching costs.
- 4 Quality is the critical order qualifier: FDA compliance is existential (2022 Sturgis crisis), but quality alone does not differentiate Abbott from Enfamil or Gerber.
- 5 The supply chain directly enables the business strategy - Lean & Six Sigma discipline drives the operational excellence behind delivery reliability and quality, while aseptic manufacturing, global facility placement, and cross-portfolio margin management translate strategy into competitive advantage no pure-play competitor can replicate.

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Appendix

- 2025 Financial Overview
- Key Competitors
- Supply Chain Flow Overview
- Manufacturing Footprint
- Key Manufacturing Facilities
- Wic Overview
- Regulatory Risk: FTC WIC Collusion Investigation

2025 Financial Overview

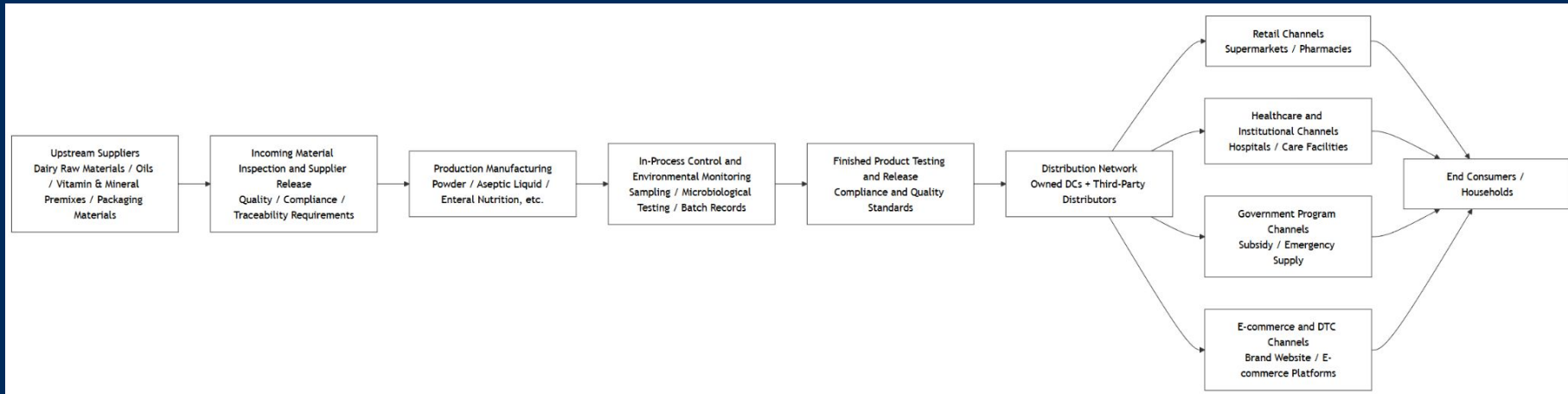
(in millions)	U.S	International	Total
Pediatric Nutrition	2,158	1,816	3,974
Adult Nutrition	1,448	3,029	4,477
Total	3,606	4,845	8,451

Source: Abbott Laboratories, Form 10-K, Fiscal Year Ended December 31, 2025.

Key Competitors

Competitors	Positioning
Mead Johnson	Infant formula; strong U.S retail
Nestle	Broad nutrition portfolio; strong R&D but less U.S hospital penetration
Danone (Nutricia)	European clinical nutrition leader; limited U.S consumer presence
Perrigo	Price-competitive store brand; no clinical positioning
Fresenius Kabi	Global leader in parenteral nutrition for critically ill patients

Supply Chain Flow Overview



Manufacturing Footprint

Reportable Segments	Manufacturing Sites
Medical Devices	33
Diagnostic Products	21
Established Pharmaceutical Products	22
Nutritional Products	13
Worldwide Total	89

Key Manufacturing Facilities

Country/Region	Site/Location
United States	Sturgis
United States	Columbus
United States	Casa Grande
United States	Tipp City
United States	Bowling Green
Ireland	Cootehill
Spain	Granada
Singapore	Singapore
Netherlands	Zwolle
China	Jiaxing
India	Jhagadia

WIC Overview

How WIC Rebates Work:

1. State awards multi-year sole-source contract based on highest rebate bid.
2. WIC participants purchase contract brand at authorized stores using EBT card.
3. State reimburses retailer at full retail price.
4. State bills manufacturer for the rebate offered in winning bid.

U.S. infants covered by WIC	39% of all births
Share of U.S formula sold through WIC	>50%
Contracts per state	1 for standard formula
Manufacturers holding nearly all contracts	3 (Abbott, Reckitt, Nestle)

Sources: National WIC Association (2019); GAO-25-106503; National Academies Press (2024); FDA Medical Foods FAQ

Risk: FTC WIC Collusion investigation

Relevant to Price (P) : Order Qualifier

What	FTC investigating whether Abbott, Reckitt (Enfamil) and Nestle colluded on WIC infant formula contract bids across U.S. states.
Scale	3 manufacturers hold ~70 of 90 state WIC contracts. A highly concentrated market raising anti competitive concerns.
Status	Investigation opened 2023; FTC published findings March 2024. No charges filed as of March 2026. Outcome pending.
Strategic Risk	If found in violation, Abbott's WIC lock-in strategy. Order Winner #1 could face structural disruption or forced contract restructuring.
Order Qualifier Link	Similar rebates across all 3 firms may reflect coordination, not competition. Either way price cannot differentiate — confirming Price as Order Qualifier, not Winner
Sources	ConsumerNotice.org ABC News GMA (2023) Food & Power (Mar 2024) FTC Infant Formula Report (2024)

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